

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-010 - Lakwa-010

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000112
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	1999-07-24 / 1999-11-21
Measured / TVD	2693 m / 2661 m
Trajectory	Vertical; inc 0.3 deg; az 298.1 deg
Formation Interval	1548-2546 m
Lithology	Fine-medium sandstone with shale interbeds
Pore Pressure Reference	3176 psi
Fracture Gradient	14.85 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Moderate (54.7/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	673	Surface
9 5/8 in casing	1561	Intermediate
7 in liner/casing	2369	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
1999-08-09 486 m	Kick	Critical	Unexpected background gas rise preceded influx. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
1999-08-13 597 m	Kick	Critical	Unexpected background gas rise preceded influx. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
1999-09-08 1161 m	Torque Spike	Critical	Rotary speed fluctuated during constant pump rate. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
1999-08-05 398 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
1999-08-20 754 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
1999-08-30 965 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
1999-09-23 1489 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Use offset torque trend to set adaptive operating envelope.

Completion Summary

The synthetic well reached 2693 m MD in the Tipam interval. The NWIS knowledge record links 17 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.